

Notes | CHip. H. 2025

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Eval Criteria

Ch. 4. Model evaluation

Factual consistency

Safety

- Categories.
- Inapp. language
 - Harmful recommendation
 - discrimina - tory behaviour
 - threat & graphic violence
 - Stereotype
 - Political Biased

FB Hate Speech detection

Skolkovo insti's toxicity classifier

Perspective API

Real Toxicity Prompts

* Instruction following capabilities *

- core requirement
- InstructGPT - finetuned for following instruction
- domain specific ability doesn't necessarily make IFC.
- Hard to quantify & is $f(\text{model}, \text{instruction})$

• Instruction following criteria.

benchmark → IFEval (Google)

→ INFOBench → additionally

To verify,

they created Y/N ques.

ans. by Human or AI evaluator

GPT.4.0 is good

content constraint

linguistic guideline

Style rule

in

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• Roleplaying :

Purpose → Entertainmant
→ or to improve quality of o/p
evaluation → Hard to automate
→ Benchmark → RoleLLM
→ Character Eval

whether stays in character

style & char. knowledge part

Don't want NPC to accidentally give spoilers
include

! System instruction ??

! User prompt ??

* Cost & Latency *

- quality, latency & cost should be balanced
- many companies go for lower-q. model if have low cost & better latency.

∴ Cost & latency optimization (ch.9)

- Pareto optimization $\equiv$ Optimizing for multiple objective

① clarity on compramisable & non-compramisable obj

② for latency some are metrics → time to first token

f → time per token

→ time b/w token

→ time per query, etc

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- Latency = $f(\text{U. model, prompt, sampling variable})$
- eg. Airoregg L.M. need stopping instruction to limit latency.
- must have & nice-to-have
- hosting a model v/s use model API

* Model Selection *

Prompt engg - Bigger to smaller

Finetuning - Smaller to Bigger

Selection procedure → ① Figuring out best achievable performance
→ ② cost-performance selection

* Model Selection Workflow *

first differentiate hard attributes & soft attributes

Steps - ① filter based on hard attribute

② Narrow down based on public info, balancing diff obj such as model quality, latency, cost GLE

③ experiment with own evaluation pipeline, balancing

④ Monitoring - detect failure & collect feedback.

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* Model build v/s buy *

- Open source, open weight & model licenses

licenses MIT

Apache 2.0

GNU

RSD

Creative Commons

model distillation

- Open source models v/s model APIs

inference service

model API

→ model provider - OpenAI, Anthropic

→ cloud service provider - Azure, GCP

→ T.P. api provider - Databricks, Mosaic, Anyscale etc

models data lineage

* Host model or use API

① Data privacy

② Data lineage and copyright

③ Performance MMLU Benchmark?

④ Functionality - Scalability, function calling, str. outputs,

f o/p guardrails,

⑤ API cost v/s Engg cost

cost, community support,

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⑥ Control, access and transparency
↓
customizability

use in production, unreliability, unpredictability is problem

⑦ On-device deployment

* Navigate Public Benchmark *

Google's BIG-bench has 214 benchmark (OpenAI's eval400)

Evaluation harness - tool helps to eval a model on multi-benchmark

- Benchmark Selection and aggregation

to identify promising model for use case

- what benchmarks to include in leaderboard?

- How to aggregate these benchmark results to rank models?

- Public leaderboards

lot of public leaderboard avoid expensive model

↓
MS MARCO left HELM

HF left HumanEval

Also decision-making isn't clear to user

HF Open LLM → 6 B.M → ARC-C - (2018)

MMLU (2020)

in

HellaSwag (2019)

✓

TruthfulQA (2021)

Ⓢ

h1noGrande (2019)

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GSM-8K (2021)

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- Competitive Math - (MATH)
- Legal - Legal Bench
- Medical - MedQA
- translation - WMT 2014
- Reading comprehension - ans. que - Narrative QA & OpenBookQA
- General que answering - Natural Questions
- Benchmark correlation
- Custom leaderboards with public benchmark
 - gather list of benchmark that eval capability as applⁿ
 - Reliability of benchmark
- Data contamination with public benchmark
- How data contamination happens
 - previous benchmarking data goes to training newer ^{model}
- Handling data contamination
 - detect contamination & decontaminate that data
 - 1) N-gram overlapping
 - 2) Perplexity

①

②

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* Design Your Evaluation Pipeline *

Step 1: Evaluate All components in a system
diff level eval eg. per task, per turn, per interm. o/p.
multiple steps & msgs.

ex. of task based eval — twenty-question

Step 2: Create an evaluation guideline

we should know ~~how~~ what bad response look like.

- Define evaluation criteria

A correct response is not always a good response

3 criteria — ① Relevance

② Factual consistency

③ Safety

- Create scoring rubrics with example.

validate with humans

clear guideline is the backbone of reliable eval pipeline

- Tie evaluation metrics to business metrics

eg. Factual consistency of 80% : we can automate
30% of customer support requests.

Understand impact of e-metrics on B-metrics in
daily, weekly, monthly active user — stickiness metrics
conversation/month or duration — engagement metrics

Step 3: Define evaluation methods and Data

• Select evaluation method

Logprobs — how confident a model is abt gen. token

Don't be afraid to fall back on human evaluation.

• Annotate evaluation data

A finer-grained understanding of your sys can serve many-
purpose → ① Avoid potential biases

② Debug

③ Find areas for application improvement

④ Avoid falling for Simpson's paradox

• Evaluate your evaluation pipeline

getting you right signals?

How reliable is pipeline?

• Iterate